

NF- κ B at the crossroads of life and death

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The choice between life and death is one of the major events in regulation of the immune system. T cells that specifically recognize viral or bacterial antigens are selected to survive and proliferate in response to infection, whereas those that are self-reactive are eliminated via apoptosis. Even the survival of alloreactive T cells requires their proper costimulation and, when infection subsides, the activated T cells are eliminated. A major regulator of such life or death decisions is the transcription factor NF- κ B. However, NF- κ B cannot function alone. A variety of mechanisms exist to modulate its activity and thereby affect the ultimate outcome of a cell's fate.

The DNA-binding activity of NF- κ B is rapidly induced in all cell types in response to proinflammatory cytokines and the byproducts of microbial and viral infections. It was therefore anticipated that NF- κ B-induced transcription would play a central role in host defense and inflammatory responses. These predictions were bolstered by the identification of NF- κ B-binding sites in the promoter regions of genes encoding a plethora of cytokines, chemokines, adhesion molecules and enzymes that produce secondary inflammatory mediators^{1,2}. The generation of mouse models that show defective NF- κ B activation or lack essential NF- κ B subunits, such as RelA (also known as p65), has proven the importance of NF- κ B in innate and adaptive immunity, inflammation and lymphoid organ development^{3–7}. The analysis of NF- κ B-deficient mice and cells also led to the identification of another function of this versatile transcription factor: the inhibition of apoptosis^{8–11}. As we shall argue here, the ability of NF- κ B to prevent the induction of programmed cell death by a variety of proinflammatory and innate immune stimuli is tightly linked to its essential proinflammatory activity.

Rather than dwell on the aspects of NF- κ B regulation and function that have been reviewed already¹², we will focus here on insights into the anti-apoptotic function of NF- κ B and its regulation. NF- κ B is a collective term referring to dimeric transcription factors that belong to the Rel family and are regulated via shuttling from the cytoplasm to the nucleus in response to cell stimulation¹³. Mammals express five Rel (NF- κ B) proteins that belong to two classes. The first class includes RelA, c-Rel and RelB, proteins that are synthesized as mature products and do not require proteolytic processing. The second group is encoded by the

Nfkb1 and *Nfkb2* genes, whose products are first synthesized as large precursors, p105 and p100, respectively, that require proteolytic processing to produce the mature p50 and p52 NF- κ B proteins. NF- κ B dimers containing RelA or c-Rel, are held in the cytoplasm through interaction with specific inhibitors, the I κ Bs. The I κ Bs undergo rapid ubiquitin-dependent degradation after exposure to a variety of agonists, which activate the I κ B kinase (IKK) complex¹².

IKK is composed of two catalytic subunits, IKK α and IKK β , which can directly phosphorylate I κ B, and a regulatory subunit, IKK γ (also known as NEMO), whose integrity is required for the activation of this pathway (Fig. 1). At least one NF- κ B dimer, formed between RelB and p52, is subject to a different form of regulation. In the cytoplasm of nonstimulated cells, RelB is associated with the NF- κ B2 p100 polypeptide whose I κ B-like COOH-terminus is degraded after cell stimulation to release RelB-p52 dimers that translocate to the nucleus¹⁴. Activation of this process is dependent on one of the IKK subunits, IKK α ¹⁵, whereas activation of the canonical NF- κ B pathway is mostly dependent on IKK β ^{16–18}. Whereas the canonical NF- κ B pathway, which is based on I κ B degradation, is essential for innate immunity⁷, the second pathway, which is based on NF- κ B2 processing, is mostly involved in lymphoid organ development and adaptive immunity¹⁵. Once in the nucleus, NF- κ B dimers are subject to further regulation mainly through phosphorylation of the Rel proteins, which is required for full induction of NF- κ B target genes^{19,20}. Several signaling pathways, which induce the phosphatidylinositol 3 kinase (PI3K)→Akt pathway, are thought to be involved in this process²¹.

NF- κ B as an anti-apoptotic transcription factor

The first indication that suppression of apoptosis is an important NF- κ B function came from the analysis of RelA^{−/−} mice, which died at embryonic day 15 as a result of extensive liver apoptosis²². Further experiments showed that increased sensitivity to pro-apoptotic stimuli is not unique to the liver, but is also shown by RelA^{−/−} fibroblasts treated with tumor necrosis factor- α (TNF- α), which—despite its reputation—is a poor inducer of apoptosis unless accompanied by inhibitors of new RNA or protein synthesis^{8,10}. In cells that lack NF- κ B, TNF- α can induce apoptosis on its own. Other experiments, which dissected the different signaling pathways activated by TNF- α via its type 1 receptor (TNFR1), provided independent evidence for the important anti-apoptotic function of NF- κ B¹¹. Collectively, these findings show that the products of certain NF- κ B target genes inhibit apoptosis (see below). In addition, the activation of NF- κ B by TNF- α attenuates the pro-apoptotic activity of the latter. Interestingly, the apoptotic signaling cascade triggered by TNF- α (Fig. 2)—which depends on recruitment of the Fas-associated death domain (FADD) protein to the cytoplasmic domain of TNFR1 leading to activation of caspase-8—is similar to the

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cascade triggered by another member of the TNF family, Fas ligand (FasL). Unlike TNF- α , however, FasL is a poor NF- κ B activator and therefore its ability to induce apoptosis does not depend on concurrent inhibition of new gene expression²³. By virtue of their ability to activate NF- κ B and caspase-8 (or caspase-10), members of the TNF family can be regarded as either pro- or anti-apoptotic.

The suppression of apoptosis by NF- κ B is an important component of TNF- α biology. The mid-gestational lethality and massive liver apoptosis of RelA^{-/-} mice—which are also shown by mice that lack IKK β or IKK γ ^{16,18,24}—is completely suppressed by the absence of TNFR1 or TNF- α ^{4,25}. The complete dependence of the fetal liver on NF- κ B for its survival is due to the production of large amounts of TNF- α —even in the absence of infection—by the hematopoietic progenitors that populate the fetal liver²⁶. Although it is not required for liver development *per se*, NF- κ B is also important for the survival of adult liver after encounter with infectious organisms such as *Listeria monocytogenes*²⁷. NF- κ B activity also aids the long-term survival of lymphoid cells²⁸, whose demise in the absence of NF- κ B can be brought about by TNF- α ⁷ or via other mechanisms (see below). In addition to the marked increase in their susceptibility to TNF- α -induced apoptosis, NF- κ B-deficient epithelial cells show increased sensitivity to a variety of DNA-damaging chemotherapeutic drugs and ionizing radiation²⁹. As many tumors, both of lymphoid and epithelial origin, show constitutively activated NF- κ B, it is likely that the anti-apoptotic function of NF- κ B represents a major obstacle to successful cancer therapy²⁹.

How NF- κ B suppresses apoptosis

The anti-apoptotic activity of NF- κ B depends on gene induction¹¹. In fact, NF- κ B induces the expression of a number of genes whose products can inhibit apoptosis; these include cellular inhibitors of apoptosis (c-IAPs), caspase-8–c-FLIP (FLICE inhibitory protein), A1 (also known as Bfl1), TNFR-associated factor 1 (TRAF1) and TRAF2. As TRAF1 and TRAF2 are adaptor proteins required for optimal NF- κ B and c-Jun kinase (Jnk) activation, their anti-apoptotic activity is most likely due to their ability to augment the activation of NF- κ B²³. The other anti-apoptotic proteins may work in a coordinated manner to block apoptosis at multiple steps along the apoptotic signaling cascade (Fig. 3).

The best studied NF- κ B-induced anti-apoptotic proteins are the c-IAPs, which directly bind and inhibit effector caspases, such as caspase-3 and caspase-7, as well as prevent activation of pro-caspase-6 and pro-caspase-9³⁰. Thus, c-IAPs can inhibit apoptosis induced by both death receptors and mitochondria-dependent pathways. The involvement of c-IAPs in the anti-apoptotic activity of NF- κ B was shown by the

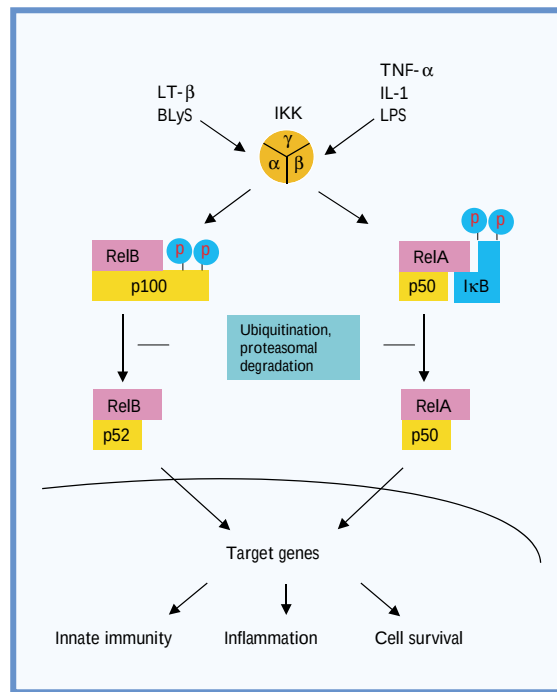


Figure 1. IKK can activate NF- κ B transcription factors via two distinct pathways. The canonical NF- κ B activation pathway—triggered by TNF- α , IL-1 or byproducts of bacterial and viral infections (such as LPS and double-stranded RNA)—is dependent on the IKK β catalytic subunit and is accomplished through I κ B phosphorylation and ubiquitin-dependent degradation. The second pathway is triggered by only a few members of the TNF family, such as lymphotoxin β (LT- β) and BLYS (also known as BAFF). This pathway is dependent on IKK α and is initiated via phosphorylation-dependent processing of p100. P, phosphate.

finding that the induction of *c-Iap2* by TNF- α or phorbol myristol acetate (PMA) + ionomycin is completely blocked in cells that stably express a degradation-resistant form (super-repressor) of I κ B α ³¹. Indeed, two functional κ B sites were found in the *clap2* promoter³². c-IAP1 expression also seems to be NF- κ B-dependent³³. The induction of c-IAP1 and c-IAP2 by NF- κ B, along with TRAF1 and TRAF2, suppresses TNF- α -mediated killing through the inhibition of caspase-8³³. Curiously, expression of c-IAP1 or c-IAP2 alone was sufficient to inhibit apoptosis induced by the DNA topoisomerase inhibitor etoposide (also a DNA-damaging agent), but did not reduce the sensitivity to TNF- α ³³. Although c-IAPs cannot bind directly to caspase-8³⁰, their recruitment to the TNFR1 signaling complex through an interaction with TRAF2³⁴ might provide sufficient proximity for inhibition³³. c-IAPs were also suggested to function through stimulation of NF- κ B activity³¹. However, this effect may be indirect and could simply be due to inhibition of caspases, which otherwise terminate NF- κ B activity (see below).

Another c-IAP regulated by NF- κ B is X chromosome-linked IAP (XIAP, also known as hILP)³⁵. XIAP inhibits caspase-3 and caspase-7 through its second baculoviral IAP repeat (BIR) domain and NH₂-terminal linker³⁶ and might prevent activation of pro-caspase-9 through a region containing its third BIR domain, BIR3³⁷. Structural analysis showed that the NH₂-terminal linker directly blocks the catalytic cleft of caspase-3 and caspase-7, whereas the BIR2 region facilitates caspase binding^{38–40}. T cell-specific transgenic expression of XIAP in mice suppressed thymocyte apoptosis induced by various death insults⁴¹. However, disruption of the gene encoding XIAP in mice showed no difference in caspase-mediated apoptosis, although c-IAP1 and c-IAP2 were up-regulated⁴². This suggests the existence of a compensatory mechanism. Overexpression of XIAP can inhibit TNF- α -induced apoptosis of cells expressing the I κ B α super-repressor mutant⁴³. These findings indicate that XIAP may be a NF- κ B target gene involved in mediating NF- κ B anti-apoptotic effects. As shall be discussed below, XIAP may also function as an inhibitor of Jnk activation⁴⁴.

Another NF- κ B-regulated inhibitor of apoptosis is c-FLIP, which was first identified as a cellular homolog of viral FLIP⁴⁵. c-FLIP contains two death effector domains (DEDs) and a catalytically inactive caspase-like domain; it can interact with FADD and pro-caspase-8 through homotypic DED-mediated interactions⁴⁶. c-FLIP may, therefore, inhibit apoptosis by interfering with pro-caspase-8 activation. Indeed, embryonic fibroblast cells from c-FLIP-deficient mice are hypersensitive to TNF- α or FasL-induced apoptosis⁴⁷. c-FLIP also interacts with TRAF2 and receptor-interacting protein (RIP)⁴⁸, which are responsible for activation of Jnk and IKK via the TNFR1 complex²³. This raises the possibility that the anti-apoptotic effect of c-FLIP might

also be due to increases in NF- κ B activation. However, TNF- α -induced NF- κ B activation was unchanged in c-FLIP-deficient cells⁴⁷. c-FLIP expression is regulated by NF- κ B, as its induction by TNF- α or PMA + ionomycin is inhibited in cells that express the I κ B α super-repressor⁴⁹. Forced expression of c-FLIP in such cells prevents TNF- α -induced apoptosis⁴⁹. Although c-FLIP is still induced by TNF- α in RelA^{-/-} cells⁴⁷, NF- κ B activation is required, albeit not sufficient, to induce c-FLIP⁴⁹. It remains to be determined whether there is a κ B site(s) in the c-FLIP promoter (encoded by *Cflar*), and if so, which NF- κ B dimers control its expression.

NF- κ B also inhibits apoptosis by DNA damaging agents, which act *via* the mitochondria-dependent pathway²⁹. This activity could be mediated through members of the Bcl-2 family, such as A1 and Bcl-x_L. A1 was cloned as a hematopoietic-specific Bcl-2 homolog⁵⁰. Its overexpression suppresses apoptosis induced by etoposide or interleukin 3 (IL-3) withdrawal and partially suppresses TNF- α -induced killing⁵¹. A1 also prevents mitochondrial depolarization, the release of cytochrome c and caspase-9 activation⁵¹. A1 may therefore exert its anti-apoptotic effects by regulating mitochondrial events. Expression of A1 is induced by several NF- κ B activators, including TNF- α , PMA + ionomycin, lipopolysaccharide (LPS) antigens or overexpression of RelA, in an NF- κ B-dependent manner^{52–54}. Overexpressed A1 can rescue c-Rel-deficient cells⁵² or I κ B α super-repressor-expressing cells⁵⁴ from apoptosis induced by TNF- α or antigen receptor ligation. The *Bfl* promoter (for A1) contains a functional κ B site^{52,54}. So far it has only been determined that A1^{-/-} mice show increased neutrophil apoptosis⁵⁵, which suggests that other genes mediate the anti-apoptotic activity of NF- κ B in other sites. One of these may be the gene encoding Bcl-x_L, which is induced in response to either CD28 or CD40 ligation, incubation with PMA + ionomycin or TNF- α and expression of the HTLV1 (human T cell leukemia virus 1) Tax protein^{53,56–59}. Overexpression of Bcl-x_L also rescues I κ B α super-repressor-expressing cells from TNF- α -induced apoptosis⁵⁹. At this point it is not clear whether Bcl-x_L is coregulated with A1 or is used to block apoptosis in different cell types or under different circumstances. These questions should be answered through the analysis of Bcl-x_L-deficient mice. It is also possible that Bcl-2 itself may be involved in the anti-apoptotic activity of NF- κ B, as its expression is defective in B cells that lack both c-Rel and RelA, the major activating subunits of NF- κ B⁶⁰. In addition, a *Bcl2* transgene partially prevents the apoptosis of such B cells engrafted to irradiated hosts⁶⁰. However, it remains to be determined whether *Bcl2* expression is directly regulated by NF- κ B.

The regulation of Bcl-2 family proteins by NF- κ B also extends to the pro-apoptotic member Bax⁶¹. Bax expression is increased in certain cells that express the I κ B α super-repressor and overexpression of NF- κ B inhibits p53-stimulated *Bax* promoter activity⁶¹. Although there is a

functional κ B site that binds RelA (p50) or RelB in the *Bax* promoter, deletion of this κ B site has no effect on NF- κ B-mediated inhibition⁶¹, which suggests an indirect effect. Importantly, TNF- α is unable to decrease Bax expression, as the half-life of Bax is very long. It is therefore postulated that NF- κ B-mediated inhibition of Bax expression may play a role only in the survival of cancer cells that show constitutive NF- κ B activity⁶¹.

It has been reported that the putative inhibitor of apoptosis, IEX-1L, is encoded by an NF- κ B target gene. Cloned by mRNA differential display as a variant of the immediate-early radiation response gene that encodes IEX-1, IEX-1L overexpression inhibits TNF- α - and FasL-induced apoptosis of Jurkat cells⁶². It was suggested that IEX-1L might be regulated by NF- κ B because TNF- α failed to induce its expression in RelA^{-/-} cells and its ectopic overexpression rescued I κ B α super-repressor-expressing cells from TNF- α -induced killing⁶². However, it may be that IEX-1L is an artificially generated variant of IEX-1 that

inhibits apoptosis by interfering with the pro-apoptotic effect of IEX-1⁶³. The existence of IEX-1L certainly needs to be verified. In addition, the biochemical functions of either IEX-1 or IEX-1L are unknown.

NF- κ B can also suppress cell death by modulating the activity of anti-apoptotic and pro-apoptotic signaling pathways. Both TRAF1 and TRAF2 are NF- κ B-inducible and contribute to its anti-apoptotic activity³³. TRAF2 plays a critical role in activating the IKK and Jnk pathways that lead to NF- κ B and AP-1 activation, respectively²³. Gene-disruption experiments showed that TRAF2^{-/-} cells are completely defective in TNF- α -stimulated Jnk activity, but only partially deficient in NF- κ B activation⁶⁴. Nevertheless, TRAF2^{-/-} mice show increased sensitivity to TNF- α -induced apoptosis⁶⁴. It is not clear whether the increased sensitivity to TNF- α is due to the partial reduction in NF- κ B activity or the

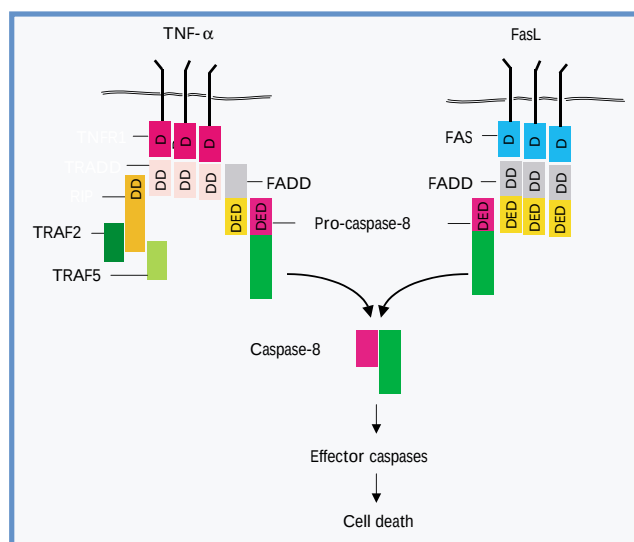


Figure 2. Signaling cascades used by TNFR1 and Fas to trigger apoptosis. The recruitment of pro-caspase-8 to Fas, TNFR1 or other death receptors upon their ligation results in its self-cleavage and activation through an induced proximity mechanism. Active caspase-8 cleaves and activates other pro-caspases. The active effector caspases cleave a variety of cellular substrates that bring about apoptotic cell death. DD, death domain; DED, death effector domain.

complete defect in Jnk activation.

The involvement of Jnk in TNF- α -induced apoptosis has been a highly controversial issue. Initial experiments based on overexpression of either Jnk activators or dominant-negative forms of the Jnk substrate c-Jun supported a pro-apoptotic role for Jnk in TNF- α signaling⁶⁵. However, the dissection of TNF- α -stimulated effector pathways failed to substantiate a pro-apoptotic role for Jnk, while identifying the strong anti-apoptotic activity of NF- κ B¹¹. Another group who used similar experiments even suggested a weak anti-apoptotic function for Jnk alongside the strong anti-apoptotic effect of NF- κ B⁶⁶. The analysis of Jnk-deficient mouse fibroblasts failed to identify any resistance to FasL- or TNF- α -induced apoptosis⁶⁷. Most likely, Jnk activation is not obligatory for TNF- α - or FasL-induced apoptosis, although it is important for ultraviolet-induced apoptosis, which proceeds *via* the mitochondria-dependent pathway⁶⁷. Nevertheless, Jnk may be a positive modulator of TNF- α -induced apoptosis *via* an unknown mechanism⁴⁴. A constitutively activated Jnk construct enhanced TNF- α -induced

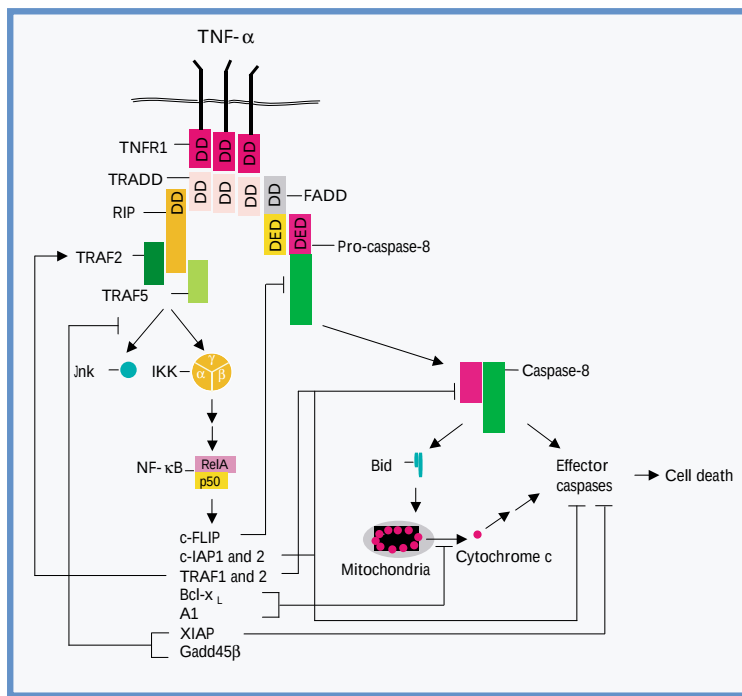


Figure 3. NF- κ B induces a variety of anti-apoptotic factors that can prevent TNF- α -induced apoptosis. These anti-apoptotic factors include inhibitors of caspase activation (c-FLIP) and action (c-IAPs), anti-apoptotic Bcl2 family members (A1, Bcl- x_L) and inhibitors of Jnk activation (XIAP, Gadd45 β).

apoptosis of RelA^{-/-} cells, whereas it had no apoptotic activity on its own⁴⁴. In addition, Jnk activation by TNF- α was highly transient in normal cells, but was extensively prolonged in IKK β - or RelA-deficient cells⁴⁴. This observation led three different groups to the discovery that NF- κ B can induce the expression of a Jnk inhibitor^{44,68,69}, which raises the possibility that this inhibitor may contribute to the anti-apoptotic function of NF- κ B.

The identity of the Jnk inhibitor is a matter of debate. By examining the NF- κ B-inducible anti-apoptotic factors discussed above, including cIAP1, XIAP, c-Flip, Bcl- x_L and Bcl-2, it was shown that only ectopic expression of XIAP was capable of restoring transient Jnk activation to TNF- α -treated RelA^{-/-} cells⁴⁴. XIAP had no effect on p38 or extracellular signal-regulated kinase (Erk) activation and its induction was indeed defective in RelA^{-/-} cells⁴⁴. Interestingly, the kinetics of Jnk activation by IL-1 were not altered by the loss of NF- κ B activity, which suggests that XIAP or other NF- κ B-induced proteins (because XIAP may not be the only inhibitor of Jnk activation induced by NF- κ B) specifically interfere with signaling from TNFR1 to the Jnk cascade. A different and elegant approach was undertaken when a cDNA library from TNF- α -treated wild-type mouse fibroblasts was screened for genes capable of preventing TNF- α -induced apoptosis of RelA^{-/-} cells⁶⁸. In addition to identifying known inhibitors of TNF- α -induced cell killing, including RelA and c-FLIP, this screen netted a cDNA encoding GADD45 β ⁶⁸. Like other members of its family, expression of the GADD45 β protein is induced in response to growth inhibition or DNA damage⁷⁰. *Gadd45b* is the only gene in this family that appears to be regulated by NF- κ B, and most importantly its ectopic expression in RelA^{-/-} fibroblasts completely suppresses TNF- α -induced apoptosis⁶⁸. Curiously, the anti-apoptotic activity of GADD45 β extends to the mitochondria-dependent pathway, as it also

blocks cell death induced by DNA-damaging agents⁶⁸. Expression of GADD45 β also abrogated the prolonged Jnk activation response in TNF- α -treated RelA^{-/-} cells⁶⁸. In addition, it has also been suggested that the anti-apoptotic activity of GADD45 β may be due to its ability to inhibit Jnk activation⁶⁸. This suggestion, however, is inconsistent with most of the findings discussed above, according to which Jnk is not a critical mediator of death receptor-induced apoptosis. In addition, *Gadd45 β* ^{-/-} mice are viable and do not show the severe liver apoptosis associated with loss of NF- κ B activity (A. Fornace and D. Liebermann, personal communication).

A possible explanation for some of the results obtained^{44,68} is that GADD45 β and XIAP are not directly involved in the regulation of Jnk activity. It has been suggested that caspase activation by death receptors may result in prolonged Jnk activation⁷¹. Thus, any factor that inhibits TNF- α -induced apoptosis may obliterate the delayed phase of the Jnk activation response seen in TNF- α -treated RelA^{-/-} cells. Nevertheless, XIAP may be a genuine inhibitor of Jnk activation, as it was capable of inhibiting Jnk activation in HeLa cells, which, unlike RelA^{-/-} fibroblasts, do not undergo apoptosis after challenge with TNF- α alone⁴⁴.

The take-home lesson from the results reviewed above is that NF- κ B activates multiple target genes whose products can block the apoptotic program triggered by either death receptors or the mitochondrial pathway. The relative importance of these factors may depend on the cell type and the particular situation in which they are examined. Strangely, none of the mouse strains analyzed so far that lack a single NF- κ B-inducible anti-apoptotic factor have shown the extensive liver apoptosis associated with loss of RelA or IKK β . Either multiple NF- κ B-induced factors are involved in the suppression of TNF- α -induced apoptosis or the critical factor is yet to be identified.

Pro-apoptotic stimuli and NF- κ B activation

If NF- κ B-dependent gene expression is such a major roadblock on the path to apoptosis, then pro-apoptotic stimuli must find ways to circumvent it. Indeed, several key components in the NF- κ B activation pathway seem to be targeted by caspases, which results in the termination of its anti-apoptotic activity.

At least two of the signaling proteins involved in TNF- α -induced NF- κ B activation, RIP and TRAF2, are caspase substrates. Caspase-8 cleaves RIP at Asp³² (ref. 72). This proteolysis, which produces an NH₂-terminal truncated fragment (RIPc) and a COOH-terminal-truncated fragment (RIPn), eliminates the ability of RIP to signal to IKK⁷². Overexpression of RIPc enhances the association between TNFR1, TRADD and FADD, while inhibiting NF- κ B activation, thereby promoting cell killing by TNF- α ⁷². TRAF2 is also proteolyzed or sequestered into a nonsoluble cellular compartment upon recruitment to CD30 (a member of the TNFR family), which results in increased sensitivity to TNF- α ^{73,74}. This down-regulation of TRAF2 appears to be specific to CD30 ligation^{75,76}. Proteolysis of TRAF1 may also contribute to turning off of the anti-apoptotic activity of NF- κ B^{76,77}. TRAF1 is cleaved at Asp¹⁶³ by caspase-8 during TNFR1- or Fas-induced apoptosis, which generates two fragments^{76,77}. Expression of the TRAF1 COOH-terminal fragment inhibits NF- κ B activation by cotransfected TRAF2 or TNFR1 and promotes TNF- α - and Fas-mediated killing^{76,77}. As the COOH-terminal fragment contains the TRAF domain, it may disrupt the interaction of TRAF1 with TRAF2, as well as with c-IAP1 and c-IAP2⁷⁷ by acting as a dominant-negative isoform.

The IKK β catalytic subunit of the IKK complex is essential for NF- κ B activation by TNF- α ¹⁸. IKK β , but not IKK α or IKK γ , is proteolyzed by caspase-3-related caspases during TNF- α - or Fas-induced apoptosis⁷⁸. Cleavage of IKK β occurs at Asp⁷⁸, Asp²¹⁴, Asp³⁷³ and Asp⁵⁴⁶ and results in the elimination of its enzymatic activity⁷⁸. One of the fragments, IKK β (1–546) appears to act as an inhibitor of IKK and can promote TNF- α -induced apoptosis upon overexpression⁷⁸. Most importantly, overexpression of a caspase-resistant form of IKK β can prevent TNF- α -induced apoptosis⁷⁸. The proteolysis of IKK β may therefore be a critical event in the decision between life and death. In wild-type cells, IKK was activated repeatedly in response to TNF- α , but eventually its activation was blunted as IKK β was proteolyzed. In contrast, in cells expressing the caspase-resistant IKK β mutant, IKK was repeatedly activated for a long time⁷⁸.

Ubiquitination-mediated proteolysis of I κ B α by the 26S proteasome is a critical step in NF- κ B activation¹². Caspase-3-mediated cleavage can prevent I κ B α degradation by separating its regulatory NH₂-terminal domain (which contains the IKK phosphorylation sites) from the body of the inhibitor. This generates a super-repressor-like molecule that is resistant to TNF- α -induced phosphorylation and degradation⁷⁹. Phosphorylation of I κ B α by IKK or substitution of Ser³² and Ser³⁶ with glutamates to mimic their phosphorylation blocked proteolysis by caspases⁸⁰. Thus, I κ B α may be another important component in the life or death switch.

NF- κ B itself is also on the caspase hit-list. RelA is cleaved during apoptosis induced by growth factor withdrawal at its COOH-terminal activation domain by several caspases⁸¹. This generates RelA with a truncated COOH-terminal that can bind DNA but is unable to activate transcription⁸¹. Thus, the proteolysis of RelA converted NF- κ B into its own inhibitor, while expression of a caspase-resistant RelA mutant can protect cells against death caused by growth factor withdrawal⁸¹.

To efficiently eliminate the protection conveyed by the IKK–NF- κ B survival pathway, apoptotic cells also use caspases to directly cleave NF- κ B-induced anti-apoptotic gene products. For instance, c-IAP1 is cleaved by caspase-3-related caspases and overexpression of c-IAP1 proteolytic fragments induces apoptosis⁸². Likewise, cleavage of XIAP by caspases at Asp²⁴ resulted in two proteolytic fragments: the NH₂-terminal fragment weakly inhibits caspase-3 and caspase-7, whereas the COOH-terminal fragment is capable of potentiating Fas-induced apoptosis³⁷. Bcl-x_L is cleaved by caspases during apoptosis induced by cytokine withdrawal, viral infection^{83,84} or by a different protease, called calpain, during apoptosis caused by ischemic injury or exposure to amyloid β peptide⁸⁵. Proteolytic cleavage converts Bcl-x_L into two pro-apoptotic fragments^{83,84}. However, it is yet to be determined whether Bcl-x_L is proteolyzed during TNF- α -induced apoptosis and whether the cleavage-resistant Bcl-x_L (D61A) mutant can suppress cell killing by TNF- α .

In addition to these caspase-dependent mechanisms, the ingestion of apoptotic bodies by macrophages seems to turn off the induction of NF- κ B-dependent target genes⁸⁶. Although the mechanism that accounts for this phenomenon remains to be explored, the uptake of apoptotic bodies down-regulates the production of TNF- α and other inflammatory cytokines and instead switches on the expression of anti-inflammatory cytokines such as transforming growth factor- β (TGF- β). This response is likely triggered through the binding of phosphatidylserine (PS)—which is present on the outside of apoptotic bodies—to the PS receptor (PSR)⁸⁷. The parasite *Trypanosoma cruzi* takes advantage of this phenomenon by inducing extensive lymphocyte apoptosis, which suppresses the inflammatory response of macrophages that ingest the apoptotic lymphocytes. This provides a

hospitable environment that promotes the replication of *T. cruzi* within macrophages⁸⁸. A different mechanism that also leads to inhibition of NF- κ B activation is used by *Yersinia pestis*, the agent that causes bubonic plague. The *Y. pestis* virulence factor, YopJ, is an ubiquitin-like protein protease that binds to mitogen-activated protein kinase (MAPK) kinase kinases (MAP3Ks) as well as IKK and inhibits their activation⁸⁹. This strategy not only allows *Y. pestis* to subdue the host inflammatory response, but likely contributes to its extreme virulence and ability to induce the apoptosis of host cells⁹⁰.

Although the majority of evidence—especially that generated by the analysis of knockout mice—provides overwhelmingly strong support to the anti-apoptotic function of NF- κ B, there are a few sporadic reports that NF- κ B may contribute to induction of pro-apoptotic molecules. This list includes death receptor 6 (DR6)—a member of the TNFR family⁹¹—DR4, DR5⁹² and Fas⁹³. However, in each case when NF- κ B-induced expression of these death receptors was detected, so was the induction of anti-apoptotic molecules that neutralized their killing activity. Thus, as yet, little physiological evidence exists for a pro-apoptotic function of NF- κ B.

Anti-apoptosis and immune regulation by NF- κ B

Because of its ability to suppress apoptosis, NF- κ B likely plays a central role in the regulation of both innate and adaptive immune responses. Although the specific elimination of virally and bacterially infected cells via apoptosis may seem to be an effective defense strategy, sealing the contents of such cells within highly condensed apoptotic bodies is an anti-inflammatory mechanism that prevents the activation of innate and adaptive immune responses. In addition to interfering with antigen presentation, the formation of an apoptotic body prevents the leakage of normal cellular constituents from infected or injured cells. The release of normal cytoplasmic proteins, such as heat-shock proteins, can result in the activation of Toll-like receptors (TLRs) and the triggering of innate immune responses^{94,95}. TLR activation also stimulates the maturation of dendritic cells, thereby increasing the efficacy of antigen presentation as well as inducing the production of various cytokines that further stimulate adaptive immunity⁹⁶.

The production of proinflammatory cytokines by dendritic cells and other types of antigen-presenting cells (APCs) is of particular importance in preventing the apoptosis of T cells that have not received proper costimulation. It is well established that T cell receptor (TCR) ligation in the absence of costimulation can result in T cell apoptosis. The need for costimulation (such as ligation of CD28 on T cells by its ligand B7 on B cells) can be bypassed with the use of adjuvants, which have long been known for their ability to potentiate the production of antibodies to T cell-dependent antigens. All adjuvants are potent TLR agonists and therefore activators of NF- κ B⁹⁴. TCR stimulation *in vivo* (by antigen injection) in the presence of adjuvants results in the induction within T cells (which do not undergo apoptosis under these conditions) of RelB and the I κ B-like protein Bcl-3⁹⁷. Unlike the conventional I κ Bs, Bcl-3 binds to p50 or p52 homodimers in the nucleus and acts as a coactivator that converts them from repressors of NF- κ B target genes to activators⁹⁸. Thus, its induction in T cells is likely to stimulate the expression of NF- κ B target genes. Indeed, the introduction of ectopic Bcl-3 to T cells prevents the induction of apoptosis by TCR ligation in the absence of costimulation or adjuvants⁹⁷.

Whereas costimulation acts directly on the T cell to potentiate IKK activation by TCR ligation, T cells do not express TLRs. Most likely, the anti-apoptotic effects of adjuvants in this system are exerted *via* TLR activation on dendritic cells. Another way in which the anti-apoptotic

activity of NF- κ B can promote adaptive immune responses is by contributing to the maturation of B cells, which is required for production of antibodies to T cell-dependent antigens¹⁵. As B cells mature, they become less sensitive to apoptosis. This process depends on the IKK α subunit of IKK, which specifically promotes the activation of p52-RelB dimers. It remains to be determined whether these dimers activate anti-apoptotic genes other than those induced by the p50-RelA dimers.

Regardless of the mechanism by which it is achieved, these examples clearly illustrate how the anti-apoptotic activity of NF- κ B is intimately linked to its ability to stimulate and orchestrate innate and adaptive host defenses. As discussed above, the ability of NF- κ B to suppress apoptosis and promote inflammation is likely mediated via a large number of target genes. One should not forget, however, that NF- κ B transcription factors do not work alone. Many NF- κ B target genes are also responsive to other transcription factors, including AP-1 and C/EBP- β . As the activities of such transcription factors are regulated by other signaling pathways, such as the MAPKs, the combinational nature of transcriptional control provides a means for modulating the repertoire of NF- κ B target genes that are induced in response to a given stimulus. Such modulation likely plays a key role in determining the exact nature of immune responses and host defenses elicited by different pathogens.

In conclusion, NF- κ B is a central regulator of innate and adaptive immune responses. This function is accomplished through the induction of genes, some of which promote inflammation, leukocyte migration and activation, whereas others act as potent inhibitors of apoptosis. It may be desirable under certain circumstances, such as during cancer therapy, to dissociate the immunoregulatory function of NF- κ B from its anti-apoptotic activity. However, this may not be easily achieved without a better understanding of all the mechanisms involved in the activation of specific and physiologically relevant NF- κ B target genes. The complete understanding of NF- κ B-dependent gene regulation is therefore a major challenge for future research.

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